MetaGraphSci: Meta-Informed Graph Transformer Self-Supervised Contrastive Neighbor Adaptation for Scientific Document Classification

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Abstract

Scientific document classification under scarce supervision is a challenging task because relevant signal is distributed across text, bibliographic metadata, and the citation graph structure. Existing text-only approaches, even when backed by powerful pretrained encoders, ignore relational and structural cues, while graph-based methods typically struggle on very small labeled sets due to over-smoothing and the absence of rich semantic priors. We present **MetaGraphSci**, a semi-supervised framework that combines a scientific pretrained language model with a *Graph Transformer* operating over the citation neighborhood and a dedicated metadata encoder. Unlike conventional message-passing networks, the Graph Transformer injects learnable structural position encodings derived from citation proximity and co-authorship into the multi-head attention mechanism, enabling attention to be simultaneously informed by semantic similarity and relational structure. Bibliographic metadata—venue, year, and author affiliation—is further used as an auxiliary conditioning signal in a *meta-informed contrastive neighbor adaptation* objective. During self-supervised pretraining, positive pairs are formed from documents sharing both structural proximity and metadata affinity, while the negative set is debiased by masking K-hop neighbors from the repulsion term. This avoids false-negative penalties between topically related documents. For fine-tuning under low-label regimes (1%–25% labeled), we adopt class-adaptive pseudo-labeling whose per-class confidence thresholds scale with the current prediction distribution, mitigating over-prediction of dominant classes. Experiments on **Cora**, **PubMed**, **ogbn-arxiv**, and **FoRC4CL** demonstrate that MetaGraphSci consistently outperforms text-only, citation-pretrained, and graph neural network baselines, with the largest gains in the most label-scarce regimes.

1 Introduction

Automatic classification of scientific documents is central to digital library management, scholarly search, and recommendation systems. Yet the task remains challenging: annotating a paper into a fine-grained research taxonomy requires domain expertise, so in practice only a small fraction of a corpus ever carries verified labels. The vast majority of documents are unlabeled, but they exist within a rich structure of citation links and bibliographic attributes.

Current approaches address this challenge from three largely separate angles. *Text-based encoders* such as SciBERT [Beltagy et al., 2019] and the citation-pretrained SPECTER [Cohan et al., 2020] and SciNCL [Ostendorff et al., 2022a] offer strong semantic representations, but they either ignore graph topology entirely or fold citation information only coarsely into the pretraining objective. *Graph-based transformers* such as GraphGPS [Rampasek et al., 2022] and NodeFormer [Wu et al., 2022] address over-smoothing in message-passing networks by replacing fixed neighborhood aggregation with global or local attention, but they do not leverage scientific metadata and have not been evaluated under the extremely low-label regimes that are common in scholarly corpora. *Weakly supervised* methods can exploit soft document-level signals but typically require carefully curated label prototypes that are unavailable in fine-grained scientific taxonomies [Meng et al., 2020].

We argue that the core limitation shared by these lines of work is the absence of a unified inductive bias that simultaneously respects (i) the *semantic geometry* encoded in pretrained language model representations, (ii) the *relational structure* of the citation graph, and (iii) the *metadata affinity* between documents sharing venue, time period, or authorship. When labeled data are extremely scarce, each of these signals can compensate for the weaknesses of the others.

To address this, we propose **MetaGraphSci**, a semi-supervised framework built around three complementary ideas.

Graph Transformer over citations. We replace the standard message-passing aggregation with a Graph Transformer [Yun et al., 2019] that attends over the citation neighborhood using multi-head attention. Crucially, each attention query is augmented with a *structural position encoding* computed from the K-hop distance in the citation graph and spectral graph features. This allows the transformer to distinguish topologically near neighbors from distant ones even when their text embeddings are similar, a distinction that flat GNN layers cannot capture without explicit depth.

Meta-informed contrastive neighbor adaptation (MCNA). We introduce a self-supervised pretraining objective in which positive pairs are constructed jointly from citation proximity and metadata affinity—two documents form a positive pair only if they are structurally adjacent *and* share at least one metadata attribute (e.g., the same venue or overlapping authors). The contrastive loss is debiased by removing K-hop neighbors from the negative pool, avoiding false-negative penalties between structurally related documents. This extends the neighborhood-aware debiasing idea with an explicit metadata gate, encouraging the encoder to treat documents sharing both structure and metadata as semantically similar. This approach is inspired by metadata-informed contrastive learning in Taechoyotin and Acuna [2024] and self-supervised citation classification in Ostendorff et al. [2022b].

Class-adaptive pseudo-labeling for low-label fine-tuning. Following Yang et al. [2025] and Chuang et al. [2024], we fine-tune the pretrained encoder with a semi-supervised objective that selects pseudo-labeled documents using per-class adaptive thresholds. This avoids the systematic bias of a fixed global threshold toward high-confidence majority classes and remains effective even when as few as 1% of training documents are labeled.

The main contributions of this work are:

1. A Graph Transformer encoder for scientific documents that incorporates structural position encodings derived from citation graph topology into the multi-head attention mechanism.

2. A meta-informed contrastive neighbor adaptation objective that uses bibliographic metadata as an auxiliary positive-pair criterion and debiases the negative set by excluding K-hop citation neighbors.

3. A two-stage training procedure combining self-supervised graph-transformer pretraining with class-adaptive pseudo-label fine-tuning, evaluated under 1%–25% labeled data regimes.

4. Empirical analysis on four benchmarks showing that the proposed combination consistently improves over text-only, citation-pretrained, and graph neural network baselines.

2 Related Work

2.1 Text-Based Scientific Document Representation

A central challenge in scientific document classification is that text-only representations often miss structure specific to scholarly corpora. Section-aware modeling has been explored in CoSAEmb [Singh and Singh, 2024], which builds contrastive embeddings from different sections of a paper to model the distinct semantic roles of introduction, methods, and results. Metadata-informed contrastive learning was introduced in MISTI [Taechoyotin and Acuna, 2024], which aligns scientific text with associated image and metadata signals to build richer cross-modal representations. These works demonstrate that scientific documents carry discriminative information well beyond the title and abstract, and that contrastive objectives are an effective vehicle for integrating it.

2.2 Semi-Supervised and Weakly Supervised Text Classification

When labeled data are scarce, semi-supervised learning requires careful strategies for utilizing unlabeled examples. Meng et al. [2020] showed that even extremely weak supervision—in the form of user-provided word sets rather than labeled documents—can bootstrap effective classifiers through a denoised self-training loop. LLM-informed semi-supervised learning [Sosea and Caragea, 2025] has more recently proposed to leverage large language models to generate soft labels for unlabeled text, which are then refined through iterative pseudo-labeling. The LPLS strategy [Chuang et al., 2024] introduced a principled approach to selecting pseudo-labeled samples in active learning by tracking the labeling status of individual instances across training epochs, reducing confirmation bias. Pseudo-label calibration with class distribution information was proposed in Yang et al. [2025] to correct the tendency of threshold-based selection to over-accept samples from majority classes. We adopt and extend this calibration idea in our class-adaptive pseudo-labeling formulation.

2.3 Citation-Informed Representations

Citation information provides a complementary view of document similarity that is not directly recoverable from text. SPECTER [Cohan et al., 2020] and SciNCL [Ostendorff et al., 2022a] incorporate citation links into pretraining contrastive objectives, producing embeddings where co-cited documents are close in representation space. Ostendorff et al. [2022b] further showed that self-supervised contrastive objectives operating on citation structure can effectively adapt pretrained language models to the citation classification task, without requiring manually labeled citation intent. Beyond direct citation links, Liang et al. [2025] demonstrated that richer inter-document relations—including finding-level and biomedical knowledge-graph edges—yield representations that generalize better than citation-only supervision. These findings motivate our decision to augment citation structure with metadata affinity when forming positive pairs.

2.4 Graph Neural Networks and Graph Transformers for NLP

Graph neural networks have been widely applied to node classification on citation graphs, but standard message-passing architectures aggregate neighborhood information uniformly and are prone to over-smoothing as depth increases, which is especially problematic under low-label conditions. Graph Transformers address this by replacing fixed aggregation with attention-based pooling over neighborhoods, allowing the model to selectively weight neighbors based on both feature similarity and learned structural relationships [Yun et al., 2019]. GraphGPS

[Rampasek et al., 2022] proposed a general and scalable framework combining local message-passing with global attention, showing that the combination consistently outperforms pure MPNN or pure Transformer designs on node and graph-level benchmarks. NodeFormer [Wu et al., 2022] introduced an efficient all-pair message-passing scheme via kernelized Gumbel-Softmax that scales to graphs with millions of nodes, making it applicable to large citation graphs without explicit neighborhood sampling. Recent work has further shown that augmenting Graph Transformer attention with explicit structural position encodings—computed from shortest-path distances, Laplacian eigenvectors, or random-walk statistics—improves node-level tasks by preserving topological context that flat attention would otherwise discard [Dwivedi and Bresson, 2022]. Our topology encoder builds on this foundation and extends it by conditioning attention on metadata compatibility between the query document and its citation neighbors, an aspect not addressed by any existing graph transformer baseline.

2.5 Field Classification Benchmarks

Benchmarks have clarified the difficulty of fine-grained scientific classification. Multi-label field classification of scientific documents using both expert-curated and crowd-sourced knowledge was studied in Gelles and Dunham [2024], highlighting the gap between coarse-grained and fine-grained taxonomies. FoRC4CL [Ahmad et al., 2024a] introduced a manually annotated benchmark of ACL Anthology papers organized under a hierarchical CL/NLP taxonomy. Subsequent shared tasks [Ahmad et al., 2024b, Francis et al., 2025] and hierarchical analyses [Dampa, 2025] showed that classification remains difficult under imbalanced and hierarchical label spaces, and that stronger systems consistently combine text, metadata, and graph signals.

3 Problem Statement

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a directed citation graph where each node $v_i \in \mathcal{V}$ corresponds to a scientific document and each edge $(v_i, v_j) \in \mathcal{E}$ indicates that document i cites document j. Each document i is described by three complementary information sources: textual content $\boldsymbol{x}_i$, bibliographic metadata $\boldsymbol{m}_i$, and the citation neighborhood $\mathcal{N}(i)$. The K-hop structural neighborhood is defined as:

$$\tilde{\mathcal{N}}^K(i) = \{j \in \mathcal{V} \setminus \{i\} \mid \mathrm{dist}_{\mathcal{G}}(i,j) \leq K\} \tag{1}$$

The target task is single-label multi-class classification over $\mathcal{Y} = \{1, \ldots, C\}$. Under low-label supervision, the training data are partitioned as:

$$\mathcal{D}_L = \{(i, y_i) \mid i \in \mathcal{V}_L\}, \quad \mathcal{D}_U = \{i \mid i \in \mathcal{V}_U\}, \quad \mathcal{V}_L \cap \mathcal{V}_U = \varnothing, \quad |\mathcal{D}_L| \ll |\mathcal{D}_U| \tag{2}$$

The model must estimate a classifier f_θ mapping the multimodal description $(\boldsymbol{x}_i, \boldsymbol{m}_i, \mathcal{N}(i))$ to a label $\hat{y}_i \in \mathcal{Y}$. Let Φ_θ denote the full encoder and let g_θ denote the classification head. The prediction process is:

$$\boldsymbol{z}_i = \Phi_\theta(\boldsymbol{x}_i, \boldsymbol{m}_i, \mathcal{N}(i)) \in \mathbb{R}^d, \quad \boldsymbol{p}_i = g_\theta(\boldsymbol{z}_i) \in [0,1]^C, \quad \hat{y}_i = \arg\max_{c \in \mathcal{Y}} p_{i,c} \tag{3}$$

The overall learning objective combines a supervised cross-entropy term on labeled documents and a semi-supervised term on unlabeled documents selected by class-adaptive pseudo-labeling:

$$\theta^* = \arg\min_\theta \left[\mathcal{L}_{\mathrm{sup}}(\theta) + \lambda\, \mathcal{L}_{\mathrm{ssl}}(\theta; \mathcal{D}_U, \mathcal{G})\right], \quad \lambda \geq 0 \tag{4}$$

where $\mathcal{L}_{\text{sup}}$ is the standard cross-entropy on $\mathcal{D}_L$ and $\mathcal{L}_{\text{ssl}}$ is defined over adaptively selected pseudo-labeled documents as described in Section 4.5.

4 Proposed Solution

MetaGraphSci encodes each document through three complementary branches: a pretrained scientific language model for textual semantics, a Deep & Cross metadata encoder for bibliographic attributes, and a Graph Transformer topology encoder for structural and relational context. The three representations are combined through a gated multimodal fusion mechanism:

$$\begin{aligned}
\boldsymbol{u}_i &= \text{MLP}_{\text{fuse}}\Big([\boldsymbol{h}_i^{(t)} \oplus \boldsymbol{h}_i^{(m)} \oplus \boldsymbol{h}_i^{(c)}]\Big) \\
\boldsymbol{g}_i &= \sigma\Big(\mathbf{W}_g[\boldsymbol{h}_i^{(t)} \oplus \boldsymbol{h}_i^{(m)} \oplus \boldsymbol{h}_i^{(c)}] + \boldsymbol{b}_g\Big) \\
\boldsymbol{z}_i &= \boldsymbol{g}_i \odot \boldsymbol{u}_i
\end{aligned} \tag{5}$$

Here $\boldsymbol{u}_i$ is the shared multimodal projection, $\boldsymbol{g}_i$ is an input-dependent gating coefficient, and $\boldsymbol{z}_i$ is the final latent representation. The gated fusion allows the model to learn, per document, how much to rely on each modality—text, metadata, or citation structure—depending on which signals are most informative for that paper.

4.1 Text Encoder

The text encoder maps the title and abstract of document i to a semantic representation using a pretrained scientific language model. The input sequence is defined as:

$$\boldsymbol{x}_i = [\texttt{CLS}] \oplus \text{title}_i \oplus [\texttt{SEP}] \oplus \text{abstract}_i \oplus [\texttt{SEP}] \tag{6}$$

A pretrained SciBERT encoder is applied to this sequence. The document embedding is taken from the final `CLS` hidden state:

$$\mathbf{H}_i^{(0)} = \mathbf{E}(\boldsymbol{x}_i) + \mathbf{P}, \quad \mathbf{H}_i^{(\ell)} = \text{SciBERT}^{(\ell)}\Big(\mathbf{H}_i^{(\ell-1)}\Big), \ \ell = 1, \dots, L, \quad \boldsymbol{h}_i^{(t)} = \mathbf{H}_i^{(L)}[0] \in \mathbb{R}^{d_t} \tag{7}$$

4.2 Metadata Encoder

The metadata encoder processes bibliographic attributes through a Deep & Cross Network. Categorical fields (venue, publisher) are embedded into trainable vectors; numerical fields (year) are normalized and projected; author embeddings are mean-pooled. The input metadata descriptor:

$$\boldsymbol{u}_i^{(m)} = \left[\bigoplus_{a \in \mathcal{C}} \boldsymbol{e}_i^{(a)} \oplus \bigoplus_{b \in \mathcal{R}} \boldsymbol{r}_i^{(b)}\right] \tag{8}$$

Cross network iteratively introduces higher-order feature interactions. For $\ell = 0, \dots, L_m - 1$, the final metadata embedding is:

$$\boldsymbol{x}_i^{(\ell+1)} = \boldsymbol{x}_i^{(0)}\Big(\boldsymbol{w}_\ell^\top \boldsymbol{x}_i^{(\ell)}\Big) + \boldsymbol{b}_\ell + \boldsymbol{x}_i^{(\ell)} \qquad \boldsymbol{h}_i^{(m)} = \text{MLP}_m\Big(\boldsymbol{x}_i^{(L_m)}\Big) \in \mathbb{R}^{d_m} \tag{9}$$

Beyond serving as a modality branch, the metadata embedding plays a second role in the MCNA pretraining objective: it defines a metadata affinity function used to refine the selection of positive pairs (see Section 4.4).

4.3 Graph Transformer Topology Encoder

Unlike a standard graph attention network, which computes attention coefficients purely from node feature similarity, the Graph Transformer topology encoder augments the attention mechanism with explicit *structural position encodings* (SPEs). This allows attention to reflect both semantic similarity and graph-topological distance, which is critical for distinguishing topically related but structurally distant papers from genuine citation neighbors.

Structural position encoding. For each document i, a structural position encoding $\boldsymbol{p}_i^{(s)} \in \mathbb{R}^{d_p}$ is computed from the citation graph. Two complementary signals are combined: (i) the normalized Laplacian eigenvectors $\boldsymbol{\phi}_i \in \mathbb{R}^k$, which encode global spectral position, and (ii) the K-hop distance profile $\boldsymbol{d}_i \in \mathbb{R}^K$, whose k-th entry counts the number of documents at hop distance k. These are projected and summed:

$$\boldsymbol{p}_i^{(s)} = \mathbf{W}_\phi \, \boldsymbol{\phi}_i + \mathbf{W}_d \, \boldsymbol{d}_i \in \mathbb{R}^{d_p} \tag{10}$$

The text embedding and structural position encoding are combined to form the token that serves as the query and key in the graph attention:

$$\tilde{\boldsymbol{h}}_i = \mathbf{W}_{\text{in}} \, \boldsymbol{h}_i^{(t)} + \boldsymbol{p}_i^{(s)} \in \mathbb{R}^{d_c} \tag{11}$$

Multi-head graph attention with metadata conditioning. Let $\mathcal{N}(i)$ denote the direct citation neighborhood of document i. For attention head r, the query, key, and value projections of document i and neighbor j are:

$$\boldsymbol{q}_i^{(r)} = \mathbf{W}_Q^{(r)} \tilde{\boldsymbol{h}}_i, \quad \boldsymbol{k}_j^{(r)} = \mathbf{W}_K^{(r)} \tilde{\boldsymbol{h}}_j, \quad \boldsymbol{v}_j^{(r)} = \mathbf{W}_V^{(r)} \tilde{\boldsymbol{h}}_j \tag{12}$$

To incorporate metadata affinity between the query document and its neighbors, we introduce a scalar metadata compatibility bias $b_{ij}^{(m)}$ computed from the dot product of their metadata embeddings:

$$b_{ij}^{(m)} = \boldsymbol{h}_i^{(m)\top} \mathbf{W}_b \, \boldsymbol{h}_j^{(m)} \tag{13}$$

The attention coefficient for head r is:

$$\alpha_{ij}^{(r)} = \frac{\exp\left(\frac{\boldsymbol{q}_i^{(r)\top} \boldsymbol{k}_j^{(r)}}{\sqrt{d_c/R}} + b_{ij}^{(m)} \right)}{\sum_{k \in \mathcal{N}(i)} \exp\left(\frac{\boldsymbol{q}_i^{(r)\top} \boldsymbol{k}_k^{(r)}}{\sqrt{d_c/R}} + b_{ik}^{(m)} \right)} \tag{14}$$

where R denotes the number of attention heads. The multi-head topology embedding is then:

$$\boldsymbol{h}_i^{(c)} = \sigma\left(\mathbf{W}_O \left[\bigoplus_{r=1}^{R} \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(r)} \, \boldsymbol{v}_j^{(r)} \right] \right) \in \mathbb{R}^{d_c} \tag{15}$$

Stacking L_T such layers with residual connections and layer normalization yields the final topology representation. The metadata compatibility bias $b_{ij}^{(m)}$ is the key innovation that distinguishes this encoder from prior Graph Transformers: it ensures that documents sharing venue, publication period, or authorship attract higher attention weight even when their text embeddings are less similar, directly exploiting the structured metadata signal during topology encoding.

4.4 Meta-Informed Contrastive Neighbor Adaptation

The pretraining objective drives the encoder toward representations where structurally and bibliographically related documents are close in latent space. We define positive pairs through a joint criterion that requires both citation proximity and metadata affinity.

Metadata affinity. Two documents i and j are metadata-compatible, written $\mu(i,j) = 1$, if they share at least one categorical metadata attribute (e.g., the same venue, overlapping authors, or publication within a two-year window). Formally:

$$\mu(i,j) = \mathbf{1}[\,\text{venue}(i) = \text{venue}(j) \;\vee\; \mathcal{A}(i) \cap \mathcal{A}(j) \neq \varnothing \;\vee\; |y_i - y_j| \leq 2\,] \tag{16}$$

where $\mathcal{A}(i)$ denotes the author set of document i and y_i denotes its publication year.

Positive pair construction. For an anchor document i, a positive view i^+ is sampled from documents in $\tilde{\mathcal{N}}^K(i)$ that also satisfy $\mu(i, i^+) = 1$. If no metadata-compatible neighbor exists, a standard data-augmented view (token dropout, synonym replacement) is used as a fallback. This policy ensures that the contrastive signal reinforces both topological and bibliographic coherence.

Debiased negative sampling. Documents in $\tilde{\mathcal{N}}^K(i)$ are excluded from the negative set of anchor i. Additionally, documents satisfying $\mu(i,j) = 1$ are soft-downweighted in the negative pool via a penalty weight $\rho \in [0,1)$. The meta-informed contrastive neighbor adaptation (MCNA) loss is:

$$\mathcal{L}_{\text{MCNA}} = -\frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \log \frac{\exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_{i^+})/\tau)}{\exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_{i^+})/\tau) + \sum_{j\in\mathcal{N}^-(i)} w_{ij}\, \exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_j)/\tau)} \tag{17}$$

where $\mathcal{N}^-(i) = \mathcal{B} \setminus \{i\} \setminus \tilde{\mathcal{N}}^K(i)$ is the debiased negative pool and the per-pair weight is:

$$w_{ij} = \begin{cases} \rho & \text{if } \mu(i,j) = 1 \\ 1 & \text{otherwise} \end{cases} \tag{18}$$

Compared to standard InfoNCE and to the neighborhood-aware debiasing of Zhu et al. [2021], MCNA further removes bibliographic false negatives. This is particularly important in scientific corpora where thematically related papers from the same venue are frequent in any mini-batch but may lack direct citation links.

4.5 Class-Adaptive Pseudo-Labeling

During semi-supervised fine-tuning, unlabeled documents whose predictions exceed a per-class confidence threshold are incorporated as pseudo-labeled training examples. Following Yang et al. [2025], the threshold for class c at fine-tuning epoch t is set proportional to the maximum predicted probability for that class across all unlabeled documents:

$$\gamma_c^{(t)} = \beta \max_{j\in\mathcal{D}_U} p_{j,c}^{(t)}, \quad \beta \in (0,1] \tag{19}$$

The selected pseudo-labeled set is:

$$\mathcal{U}^{(t)} = \left\{ j \in \mathcal{D}_U \;\middle|\; \max_c p_{j,c} \geq \gamma_{\hat{y}_j}^{(t)}, \quad \hat{y}_j = \arg\max_c p_{j,c} \right\} \tag{20}$$

The semi-supervised objective is the cross-entropy loss over the selected set:

$$\mathcal{L}_{\text{ssl}}(\theta; \mathcal{D}_U, \mathcal{G}) = -\frac{1}{|\mathcal{U}^{(t)}|} \sum_{j \in \mathcal{U}^{(t)}} \log p_{j,\hat{y}_j} \tag{21}$$

By scaling the threshold per class, this formulation avoids the failure mode of a global threshold, which systematically favors high-entropy minority classes or high-confidence majority classes depending on the degree of imbalance. This is a key concern in fine-grained scientific taxonomies where class distributions are highly skewed [Ahmad et al., 2024a, Dampa, 2025].

4.6 Classification Head

Classification is performed through a set of learnable class vectors $\{\boldsymbol{w}_c\}_{c=1}^{C}$. Both document representations and class vectors are ℓ_2-normalized to keep the decision rule consistent with the cosine similarity structure of the contrastive pretraining objective:

$$\bar{\boldsymbol{z}}_i = \frac{\boldsymbol{z}_i}{\|\boldsymbol{z}_i\|_2}, \quad \bar{\boldsymbol{w}}_c = \frac{\boldsymbol{w}_c}{\|\boldsymbol{w}_c\|_2}, \quad o_{i,c} = s\, \bar{\boldsymbol{z}}_i^{\top} \bar{\boldsymbol{w}}_c, \quad c = 1, \ldots, C \tag{22}$$

The class posterior and final prediction are:

$$\boldsymbol{p}_i = \text{softmax}(\boldsymbol{o}_i), \quad \hat{y}_i = \arg\max_{c \in \mathcal{Y}} p_{i,c} \tag{23}$$

The temperature scale $s > 0$ controls logit sharpness and is learned during fine-tuning.

MetaGraphSci Training Procedure

Two-stage training: meta-informed contrastive neighbor adaptation pretraining followed by class-adaptive pseudo-label semi-supervised fine-tuning.

Require: $\mathcal{G}$, $\mathcal{D}_L$, $\mathcal{D}_U$, hop radius K, temperature τ, metadata downweight ρ, confidence scaling β, loss weight λ
Ensure: Trained encoder Φ_θ and class vectors $\{\boldsymbol{w}_c\}_{c=1}^{C}$
Initialize θ (SciBERT, metadata encoder, Graph Transformer) and $\{\boldsymbol{w}_c\}$
Stage 1: MCNA Pretraining
for each pretraining epoch **do**
    Sample mini-batch $\mathcal{B} \subset \mathcal{V}$
    Compute SPEs $\boldsymbol{p}_i^{(s)}$ from Laplacian eigenvectors and hop-distance profiles
    Compute representations $\boldsymbol{z}_i = \Phi_\theta(\boldsymbol{x}_i, \boldsymbol{m}_i, \mathcal{N}(i))$ for $i \in \mathcal{B}$
    for each anchor $i \in \mathcal{B}$ **do**
        Sample positive i^+ from $\tilde{\mathcal{N}}^K(i)$ with $\mu(i, i^+) = 1$ (or augmented fallback)
        Exclude $\tilde{\mathcal{N}}^K(i)$ from negative pool; downweight metadata-compatible negatives by ρ
    end for
    Evaluate $\mathcal{L}_{\text{MCNA}}$ {Eq. 17}
    Update θ by minimizing $\mathcal{L}_{\text{MCNA}}$
end for
Stage 2: Semi-Supervised Fine-Tuning
for each fine-tuning epoch t **do**
    Sample $\mathcal{B}_L \subset \mathcal{D}_L$ and $\mathcal{B}_U \subset \mathcal{D}_U$
    Compute representations and class posteriors for $\mathcal{B}_L \cup \mathcal{B}_U$
    Evaluate supervised loss $\mathcal{L}_{\text{sup}}$ on $\mathcal{B}_L$
    Compute per-class thresholds $\gamma_c^{(t)}$ and select $\mathcal{U}^{(t)}$ from $\mathcal{B}_U$
    Evaluate pseudo-label loss $\mathcal{L}_{\text{ssl}}$ {Eq. 21}
    Update θ and $\{\boldsymbol{w}_c\}$ minimizing $\mathcal{L}_{\text{sup}} + \lambda\, \mathcal{L}_{\text{ssl}}$
end for
return $(\Phi_\theta, \{\boldsymbol{w}_c\}_{c=1}^{C})$

4.7 Datasets and Low-Label Protocol

The model is evaluated on four benchmarks spanning different scales, class granularities, and metadata richness. **Cora** and **PubMed** are classical citation graph benchmarks with node features derived from bag-of-words representations. **ogbn-arxiv** is a larger-scale directed citation graph of arXiv papers with 40 subject-area classes. **FoRC4CL** [Ahmad et al., 2024a] is a fine-grained benchmark of ACL Anthology papers annotated with a hierarchical CL/NLP taxonomy, making it the most challenging setting for class-imbalanced low-label evaluation.

To simulate low-label regimes, the training split is subsampled so that only 1%, 5%, 10%, or 25% of training nodes retain their labels. The remaining training documents are treated as unlabeled. Validation and test sets remain fully labeled to ensure stable evaluation across all conditions.

4.8 Baselines

Baselines are drawn from three categories. *Text-only baselines* include SciBERT [Beltagy et al., 2019] fine-tuned directly for classification. *Citation-pretrained baselines* include SPECTER [Cohan et al., 2020] and SciNCL [Ostendorff et al., 2022a]. *Graph Transformer baselines* include GraphGPS [Rampasek et al., 2022] and NodeFormer [Wu et al., 2022], both equipped with SciBERT node features, representing the strongest existing graph-based architectures for node classification. Where applicable, we also compare against the weakly supervised text classification approach of Meng et al. [2020] and the LLM-informed semi-supervised approach of Sosea and Caragea [2025]. All methods are evaluated under identical data splits and the same low-label protocol.

4.9 Evaluation Metrics

Following standard practice for imbalanced classification [Ahmad et al., 2024a, Dampa, 2025], we report Accuracy, Micro-F1, and Macro-F1. **Macro-F1** is the primary metric in fine-grained settings because it treats each class equally regardless of frequency. Precision and Recall are reported as supplementary diagnostics.

4.10 Ablation Design

The following variants are evaluated to isolate individual contributions:

- **w/o SPE**: remove structural position encodings from the Graph Transformer;
- **w/o metadata conditioning**: remove the metadata compatibility bias $b_{ij}^{(m)}$ from attention;
- **w/o metadata encoder**: remove $\boldsymbol{h}_i^{(m)}$ from the gated fusion entirely;
- **w/o gated fusion**: replace gated fusion with simple concatenation + MLP;
- **InfoNCE instead of MCNA**: replace the debiased MCNA loss with standard InfoNCE;
- **Fixed pseudo-label threshold**: replace class-adaptive thresholds with a single global γ;
- **GraphGPS instead of MetaGraphSci topology encoder**: replace the metadata-conditioned Graph Transformer with a standard GraphGPS local+global attention layer to isolate the contribution of metadata conditioning and SPEs;

5 Results and Analysis

5.1 Main Quantitative Results

The primary results are reported in a benchmark table comparing MetaGraphSci against all baselines under each label regime. An effective structure presents results for 1%, 5%, 10%, and 25% labeled data across all four datasets with Macro-F1, Micro-F1, and Accuracy as primary columns.

Table 1: Benchmark comparison across low-label regimes. This table reports Accuracy, Micro-F1, and Macro-F1 for each method and dataset.

Method	**Cora**			**PubMed**			**ogbn-arxiv**			**FoRC4CL**		
	Acc.	Mi-F1	Ma-F1	Acc.	Mi-F1	Ma-F1	Acc.	Mi-F1	Ma-F1	Acc.	Mi-F1	Ma-F1
SciBERT	–	–	–	–	–	–	–	–	–	–	–	–
SPECTER	–	–	–	–	–	–	–	–	–	–	–	–
SciNCL	–	–	–	–	–	–	–	–	–	–	–	–
GraphGPS [Rampasek et al., 2022]	–	–	–	–	–	–	–	–	–	–	–	–
NodeFormer [Wu et al., 2022]	–	–	–	–	–	–	–	–	–	–	–	–
Weakly-sup [Meng et al., 2020]	–	–	–	–	–	–	–	–	–	–	–	–
LLM-SSL [Sosea and Caragea, 2025]	–	–	–	–	–	–	–	–	–	–	–	–
MetaGraphSci	–	–	–	–	–	–	–	–	–	–	–	–

The accompanying analysis should focus on whether the advantage of MetaGraphSci is largest at 1% labeled data, where structural pretraining and pseudo-labeling contribute most, and whether it narrows as supervision increases.

5.2 Ablation Results

Table 2: Ablation study. Each row removes or replaces one component of MetaGraphSci.

Variant	**Accuracy**	**Micro-F1**	**Macro-F1**
Full MetaGraphSci	–	–	–
w/o structural position enc.	–	–	–
w/o metadata attention bias	–	–	–
w/o metadata encoder	–	–	–
w/o gated fusion	–	–	–
GraphGPS topology (no metadata bias)	–	–	–
InfoNCE instead of MCNA	–	–	–
Fixed pseudo-label threshold	–	–	–

5.3 Qualitative and Visual Analysis

The paper should include training and validation loss curves for both pretraining and fine-tuning stages, F1 comparison plots across label regimes, and embedding visualizations (t-SNE or UMAP) comparing representations produced by InfoNCE versus MCNA pretraining. If MCNA works as intended, same-venue and co-cited papers should form more coherent clusters without collapsing unrelated communities.

5.4 Error Analysis

Error analysis should focus on confusions between semantically adjacent categories, which are especially frequent in fine-grained taxonomies like FoRC4CL. A confusion matrix should identify which class pairs are most commonly confused. Errors should be categorized into failures

Figure 1: Placeholder for training curves, F1 across label regimes, and UMAP embedding comparisons between InfoNCE and MCNA pretraining.

attributable to lexical similarity between fields, sparse or noisy citation neighborhoods, and missing or incomplete metadata. Case studies comparing MetaGraphSci and text-only SciBERT predictions on the same documents can illustrate when metadata conditioning and structural attention correct or introduce classification errors.

6 Conclusion

This paper presented MetaGraphSci, a semi-supervised framework for scientific document classification that integrates pretrained language model semantics, bibliographic metadata, and citation graph structure within a Graph Transformer architecture. The key innovation is the meta-informed contrastive neighbor adaptation (MCNA) objective, which constructs positive pairs from documents that are jointly citation-proximate and metadata-compatible, and debiases the negative set by excluding both structural neighbors and metadata-similar documents.

Graph Transformer attention is further conditioned on metadata compatibility between the query and its citation neighbors through a learnable scalar bias, enabling structured bibliographic information to influence the topology encoding step. For low-label fine-tuning, class-adaptive pseudo-labeling scales the acceptance threshold per class to mitigate over-prediction of dominant categories.

The proposed approach addresses a realistic limitation of existing scientific document classification methods: none of the current text-only, citation-pretrained, or graph neural network baselines fully exploit the joint signal available from structure and metadata under scarce supervision. MetaGraphSci is designed to be the first method to systematically combine structural position encodings, metadata-conditioned graph attention, and debiased contrastive pretraining in a single framework.

Future work can explore extensions to hierarchical taxonomies, multi-label classification, and more scalable implementations of the K-hop neighborhood mask for very large citation graphs.

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